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The Synthesis and Identification of Codonopsis Pilosula Polyferose

ZHANG Xin-guo CHEN Wen-jie ZENG Yan-long MI Shi-tao WANG Qiang-lin LI Kun ZHANG Lin-lin
(Institute of life science and engineering, Lanzhou University of Technology, Lanzhou 730050, China)

Abstract: Objective: In this paper, we use codonopsis pilosula as the raw materials and synthesized an organic iron agent by extracting polysaccharide. Method: Codonopsis pilosula polysaccharide was obtained with extraction of water-alcohol method and deproteinization of Sevage method, and the content of polysaccharide was analyzed with phenol-sulfuric acid method. The complex-polysaccharide-ferric was prepared with Codonopsis pilosula polysaccharide and chloride ferric in alkaline condition and synthesis temperature was studied. The ferric content of the complex and existence form were analyzed with Phenanthroline Colorimetric method combined with dialysis. The ferric combination mode was analyzed by Infrared Spectra. Result: As a deep brownish red powder, the codonopsis pilosula polysaccharide-ferric was easy to dissolve in water. The yield of polysaccharide-ferric was 15.6% and the content of ferric was 19.9% with optimal synthesis temperature at 70℃. The free ferric has not been detected with dialysis and Phenanthroline Colorimetric method, all of the results showed the complex of co-donopsis pilosula polysaccharide-ferric was synthesized, and the structure was indicated by Infrared Spectra. Conclusion: According to this research, it is possible that codonopsis pilosula polyferose can exist as a complex form.

Key words: Codonopsis pilosula polysaccharide; Synthesis; Phenanthroline colorimetric method; Codonopsis pilosula polyferose complex

中药桑白皮导数光谱鉴别的研究

李洪娟*,代现平,马静
(滨州医学院药学院,山东 烟台 264003)

摘 要:目的:为桑白皮的鉴别提供科学依据。方法:采用紫外—可见光谱法和导数光谱法进行鉴别。结果:桑白皮的紫外—可见光谱和导数光谱具有一定的光谱特征。结论:本研究可为桑白皮的鉴别提供辅助手段。

关键词:桑白皮;鉴别;紫外—可见光谱;导数光谱
中图分类号:R284.1 **文献标识码**:A **文章编号**:1002-2392(2011)01-0081-02

桑白皮是桑科桑属植物 Morus alba L的干燥根皮,有泻肺平喘、利水消肿之功效。民间常用于消炎、利尿、解热、镇咳、祛痰等。现代药理试验表明桑白皮具有抗 HIV、抗呼吸道病毒、降压、抗肿瘤等多种的药理作用[1]。常见的桑属植物有桑、华桑、鸡桑、蒙桑、黑桑等,其枝、叶、根等均可入药。桑属植物的化学成分多类似,主要含有黄酮类化合物,包括桑酮、桑酮醇、桑根酮、桑根酮醇、桑素、桑皮根素等[2]。本实验采用导数光谱法对桑白皮进行鉴别研究,为今后该药材制定质量标准提供参考。

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作者简介:李洪娟(1981-),女,硕士,讲师,主要从事天然药物研究工作。E-mail: lhjczb@163.com
*通讯作者:李洪娟。

1 仪器与材料

TU-1901紫外可见分光光度计(北京普析通用仪器有限公司);SHB-Ⅲ循环水式多用真空泵(郑州长城科工贸有限公司);电子天平(Metter-Toledo仪器公司);集热式磁力加热搅拌器(郑州长城科工贸有限公司);电热恒温鼓风干燥箱(上海精宏实验设备有限公司)。无水乙醇、石油醚、乙酸乙酯和氯仿均为AR级;桑白皮购于药材市场。

2 方法与结果

2.1 供试液的制备

分别称取桑白皮粉末4份,每份2g,分别加入95%的乙醇、石油醚、乙酸乙酯、水各30mL,冷浸48h,滤过,滤液备用;另取桑白皮粉末4份,每份2g,分别加入95%的乙醇、石油醚、乙酸乙酯、水各30mL,回流提取2次,每次1h,过滤,滤液备用。

2.2 紫外光谱的测定

取上述各供试液适当稀释,以相应溶液为空白,在200～800nm波长范围内扫描,测定紫外光谱、一阶导数光谱和二阶导数光谱(见表 1～3和图 1～3)。

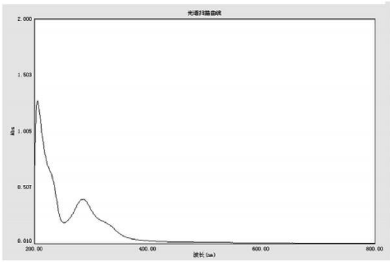

图 1 乙醇回流提取液零阶导数光谱图

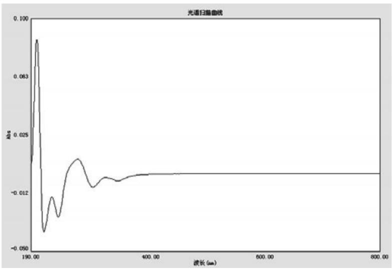

图 2 乙醇回流提取液一阶导数光谱图

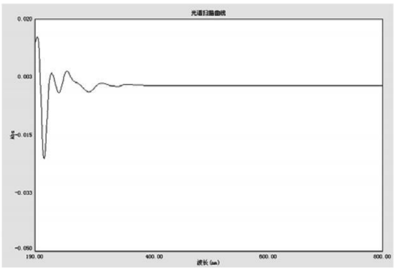

图 3 乙醇回流提取液二阶导数光谱图

表 1 桑白皮提取液零阶紫外 -可见光谱数据

溶剂	提取方式	峰位(nm)	谷位(nm)
95%的乙醇	冷浸	287, 209	252
	回流提取	286, 205	252
石油醚	冷浸	269, 238	256
	回流提取	269, 233	255
乙酸乙酯	冷浸	284, 239	252
	回流提取	283, 242	252, 224
水	冷浸	271, 200	
	回流提取	276, 200	

表 2 桑白皮提取液一阶紫外 -可见光谱数据

溶剂	提取方式	峰位(nm)	谷位(nm)
95%的乙醇	冷浸	324, 272, 226, 202	343, 300, 238, 217
	回流提取	320, 272, 227, 201	342, 298, 238, 213
石油醚	冷浸	262, 232	287, 245
	回流提取	278, 262, 228	331, 291, 240
乙酸乙酯	冷浸	316, 271	341, 295, 248
	回流提取	318, 269, 236, 215	341, 296, 248, 222
水	冷浸	266	288, 209
	回流提取	268	288, 206

表 3 桑白皮提取液二阶紫外 -可见光谱数据

溶剂	提取方式	峰位(nm)	谷位(nm)
95%的乙醇	冷浸	352, 310, 246, 222	335, 287, 232, 209
	回流提取	352, 308, 246, 220	335, 285, 233, 207
石油醚	冷浸	311, 293, 252, 225	301, 270, 239
	回流提取	342, 310, 275, 249, 222	325, 284, 267, 234
乙酸乙酯	冷浸	351, 303, 256, 217	325, 284, 242
	回流提取	352, 304, 256, 230	333, 284, 243, 218
水	冷浸	236, 215	278, 203
	回流提取	354, 295, 241, 213	250, 229, 200

3 讨论

桑白皮提取液在紫外光区都有吸收峰,特别是石油醚、乙酸乙酯、乙醇提取液的特征吸收峰较多、较强;不同提取溶剂的冷浸液、回流液特征吸收峰峰位大致相同。导数光谱是吸收度或光强度对波长变化的曲线,它能很好地解决相邻吸收带的重叠,使吸收曲线峰位移动,峰形不对称,出现肩峰等现象,使谱带宽度变小,分辨能力增高。由于紫外 -可见吸收光谱受影响的因素很多,它有一定的局限性,所以其专属性不强,只能作为鉴别的辅助手段。

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An Identification Research on Derivative Spectrum of White Mulberry Root-bark

LI Hong-juan, DAI Xian-ping, MA Jing
(Institute of Pharmacy, Binzhou Medical college, Yantai 264003, China)

Abstract: Objective: To provide the scientific evidence for the identification of white mulberry root-bark. Method: Using the method of Uv-vis spectrum and derivative spectrum. Result: The Uv-vis spectrum and derivative spectrum of white mulberry root-bark have certain spectral properties. Conclusion: The method can be used as a supplementary method in the identification of white mulberry root-bark.

Key words: White mulberry root-bark; Identification; Uv-vis spectrum; Derivative spectrum